MECHANISTIC INTERPRETABILITY NOTE

Why does a detailed rubric hurt gpt-oss-20b?

A logit-lens and causal-activation-patching investigation on BFF-Bench, following the method of Wang et al. 2025, "When Truth Is Overridden: Internal Origins of Sycophancy".

Summary. A detailed rubric raises most judges' agreement with human labels, but lowers gpt-oss-20b's specifically. Confirmed with BFF-Bench's real human-annotator rubric: **κ drops 0.667→0.617**; removing one rule — steps must match the reference, not just the answer — **fully restores it (0.666)**. gpt-oss-120b is unaffected.

Explaining this mechanistically needs a reasoning-free forced-choice probe to read internal layers — and under that setup gpt-oss-20b's effect nearly vanishes (**1 of 459 items flip**). So the main results use a broader, stricter researcher-written rubric instead: the only one that works in every model, including gpt-oss-20b, which shows a **late, switch-like computation (layer ≈22/24, 81% causal suppression)**. A secondary tab keeps the real-rubric results — strong for 4 of 5 other judges, including the study's strongest effect: **nemotron3-nano-30b's 99%/95% reversible pivot**.

1. Two rubrics

A rubric was built from scratch to probe this effect, then turned out to differ from the real one behind BFF-Bench's labels — Appendix A.4 of Setlur et al., "No Free Labels", shown below. **§2 uses the real rubric; §4's main results use ours instead** (§3 explains why).

> **Verbatim: Appendix A.4, "Annotator Guidelines," No Free Labels**
>
> 1. If the model's final answer or recommendation disagrees with the reference, it should be marked "Incorrect".
> 2. If the model agrees with the reference but its intermediate steps disagree, it should also be marked "Incorrect".
> 3. If the model agrees with the reference, all of the intermediate steps agree, and it provides the additional information that is not obviously wrong based on the reference, it should be marked "Correct".
> 4. If the model agrees with the reference, all of the intermediate steps agree, but the answer is slightly off due to rounding or performing approximate calculations, it should be marked as "Correct".
> 5. If the model agrees with the reference, all of the intermediate steps agree, but it is lacking some amount of information given within the reference, you should make a judgment call as to whether the missing information is crucial to the answer. If you are unsure, lean towards "Correct" or mark it "Unsure".
>
> *Source: arXiv:2503.05061v4, Appendix A.4 — written for human annotators, reused here verbatim as the LLM judge's grading instructions.*

RULE	OURS	APPENDIX A.4
Final answer	Match within ~1% tolerance	Must match the reference
Method / steps	Must be the standard method	Must match the reference's steps — *the only rule shared in spirit*
Completeness	Every part answered, else incorrect	*not specified*
Rounding	Still **INCORRECT** ("no partial credit")	Still **CORRECT** — protected
Extra info	*not specified*	**CORRECT** if not obviously wrong
Missing info	Counts against completeness	Judgment call; lean **CORRECT**

Ours stacks **six** strict conditions; Appendix A.4 is strict on exactly **one** (process must match) and lenient everywhere else.

2. Behavioral result

Cohen's κ, all 8 Tinker-pool judges, before/after grading with the real rubric:

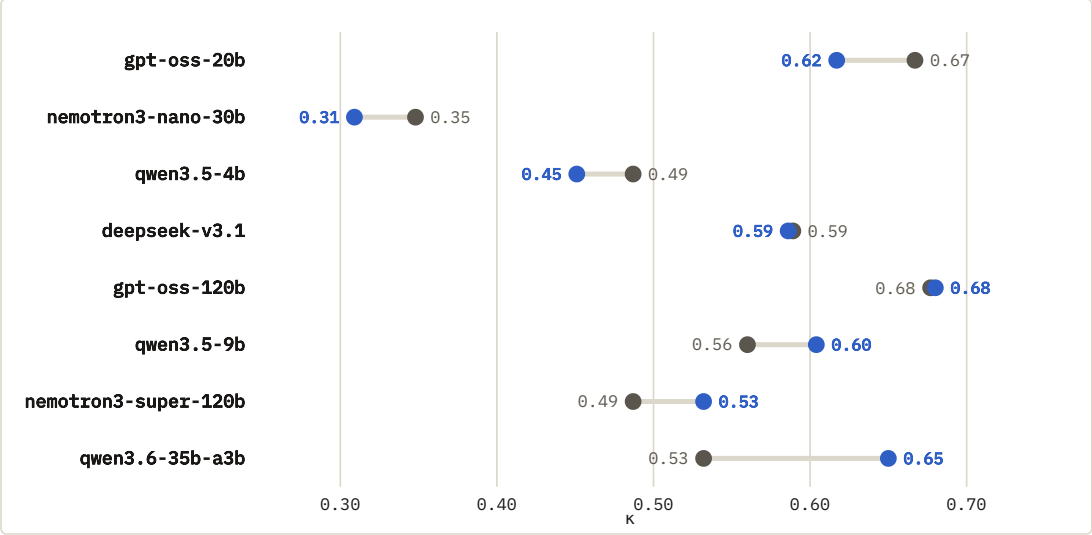


Figure 1. No rubric (gray) vs. real rubric (navy), sorted by effect size. **gpt-oss-20b** is the only judge that moves backward meaningfully; six of eight hold steady or improve.

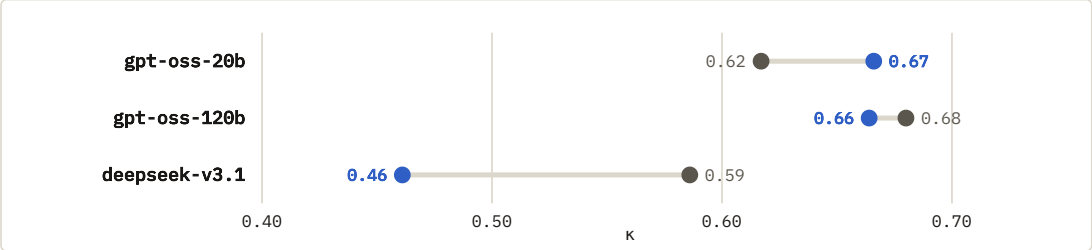


Figure 2. The paper's own ablation: real rubric (gray) vs. the same rubric minus Rule 2 (steps must match, navy). Removing it **fully restores gpt-oss-20b**; gpt-oss-120b is unchanged; deepseek-v3.1 moves the opposite way (unexplained).

▶ Exact values

TAKEAWAY 1

gpt-oss-20b is the only judge fully restored by removing one rule; gpt-oss-120b shows no effect — a 20B-CHECKPOINT PROPERTY, NOT A GPT-OSS-FAMILY ONE

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3. Why the main results use our rubric

Every interp number needs a forced-choice probe: no reasoning, an immediate CORRECT/INCORRECT read off the logits. That's a big handicap vs. how judges normally work. Accuracy with reasoning vs. without it shows how much each one depends on it:

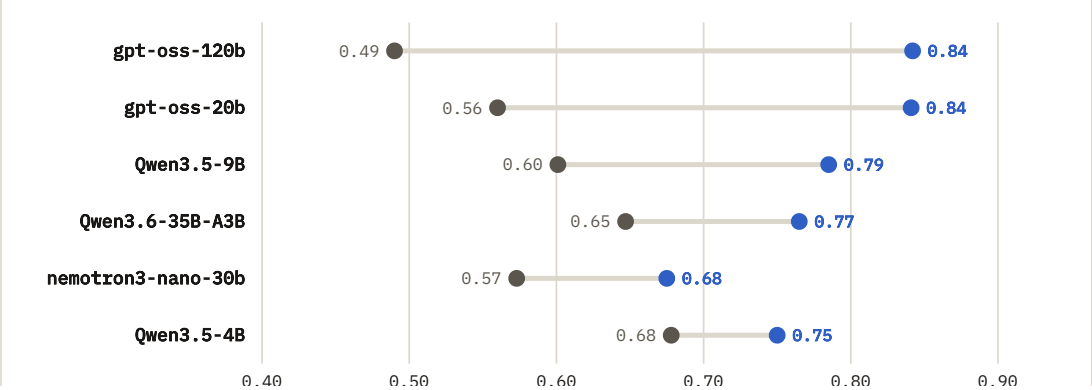




Figure C. Accuracy with reasoning (navy) vs. without it (gray), sorted by gap. The **gpt-oss family depends on reasoning most (+0.35, +0.28)** — exactly where the real-rubric probe found nothing. **nemotron3-nano-30b and Qwen3.5-4B depend on it least (+0.10, +0.07)** — and gave the clearest results. A plausible reason gpt-oss-20b resists this method: it leans on reasoning the probe doesn't allow.

The real rubric's one strict rule needs reasoning to evaluate — clean signal for 4 of 5 judges, silent for gpt-oss-20b. Our rubric stacks several bluntly-worded conditions instead of one rule requiring comparison, so it can bias an immediate token read directly, without reasoning. **That makes it more sensitive, not more faithful** — but it's the only rubric producing a flip set in every model, including gpt-oss-20b, so it's the main lens below.

Why these models

MODEL	REASON TESTED
gpt-oss-20b	The motivating case.
gpt-oss-120b	Same family, larger scale — is the effect gpt-oss-wide or 20B-specific? (latter)
Qwen3.5-9B	κ improves under rubrics — opposite direction, clean contrast.
Qwen3.5-4B	Same family as 9B, smaller scale.
nemotron3-nano-30b	Different architecture; also declines under the real rubric — closest behavioral analogue.
Qwen3.6-35B-A3B	Largest κ improvement of any judge (+0.118) — third Qwen scale point.

Not tested: nemotron3-super-120b, deepseek-v3.1 (time budget).

4. Mechanistic results

Our rubric is the main tab (per §3, it works for every model). The real Appendix A.4 rubric is secondary — works for 4 of 5 judges, not gpt-oss-20b, but includes this study's strongest causal effect where it does.

Our rubric | Appendix A.4 rubric (secondary)

Layer-wise decision tracking

At every layer, the hidden state is read through the model's own final norm and unembedding (the logit lens) to see which verdict the model would commit to if it stopped there. We track **P(CORRECT) = $p_{CORRECT}$ / ($p_{CORRECT}$ + $p_{INCORRECT}$)**, a clean 0-1 probability (the same quantity used internally for the KL divergence below), restricted to items whose verdict actually flips between prompts — averaging in items that never flip would wash out the signal.

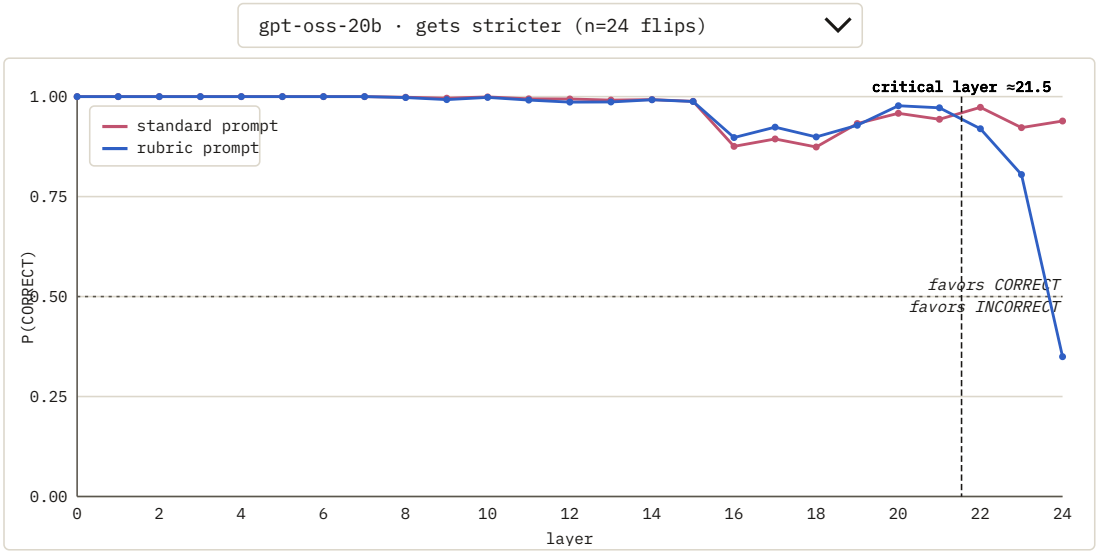


Figure 3. gpt-oss-20b, 24 flips (CORRECT→INCORRECT). Both curves favor CORRECT until a **late** reversal, just before the critical layer.

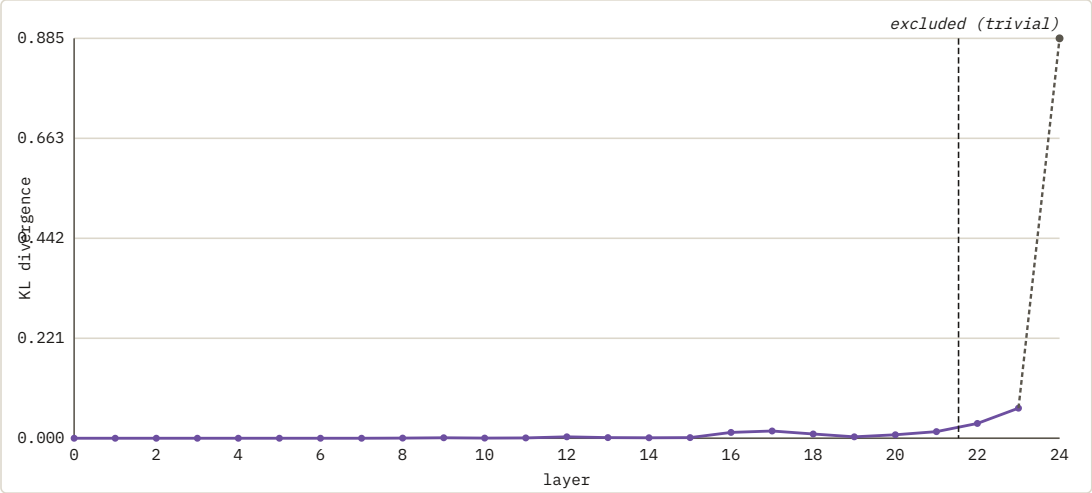


Figure 4. Layer-wise KL divergence between the standard-prompt and rubric-prompt output distributions, same flip-only item set and model selected above. The very last layer's divergence is excluded from the critical-layer search (dashed vertical line) because it reads directly off the logits and trivially dominates — shown here anyway, grayed out, so that exclusion is visible rather than hidden.

TAKEAWAY 2

GPT-OSS-20B IS ONE-SIDED

(24 stricter, ~0 lenient).

GPT-OSS-120B FLIPS BOTH WAYS ALMOST EQUALLY

(96 vs. 95) — more perturbed at the token level, but the directions cancel out, matching its small net κ change in §2.

Causal activation patching

Correlation is not causation. For every flip item, we swap the hidden state at its critical layer (3-layer window, 1 layer for the smaller Qwen patch) between conditions and re-run the model.

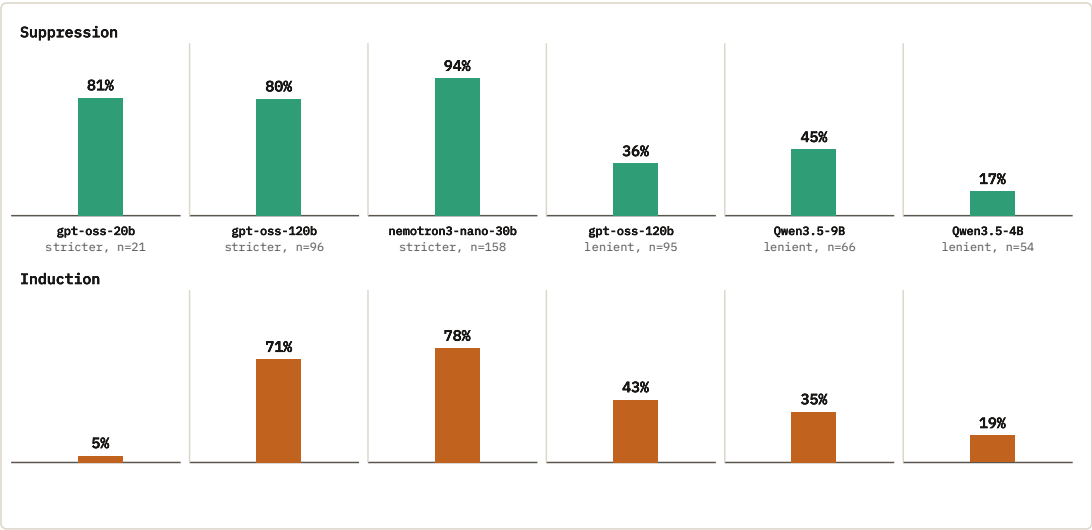


Figure 5. Our rubric throughout. **Suppression** (top): patch the better condition into the worse run, teal = share recovering the correct verdict. **Induction** (bottom): the reverse, orange = share broken. **gpt-oss-20b**: one-way gate (recovers almost always, rarely breaks). **gpt-oss-120b & nemotron3-nano-30b, stricter direction**: strongly causal *both* ways — a reversible pivot, not a gate. Lenient-direction flips (gpt-oss-120b, both Qwen) are weaker and symmetric — a softer, distributed shift. Several distinct signatures, not one mechanism at different strengths.

5. Scope of these findings

Not yet shown: a mechanism for **gpt-oss-20b's own real-rubric effect** — the model that started this investigation

> **Not yet shown:** a mechanism for **gpt-oss-20b's own real-rubric effect** — the model that started this investigation. Its κ drop (0.667→0.617) is real, but comes from the full reasoning pipeline; strip reasoning out for the forced-choice probe and the flip set collapses to **1 of 459 items**, too few to patch, even though four other judges work fine under the same setup. Fixing this likely needs the model to reason first, then be read at the final token.

Team 10 × Kensho, AC297r. Data: BFF-Bench / kensho/VERDICTS (No Free Labels, arXiv:2503.05061).
Method: Wang et al. 2025, arXiv:2508.02087.